

RISK ASSESSMENT REPORT

Organization: Silver Oak University, Ahmedabad, Gujarat

Department: MSc. (Data Science)

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1. Introduction:

Silver Oak University is a private university located in Ahmedabad, Gujarat, India. The university was recognised by the University Grants Commission (UGC) in 2019 after being declared a State Private University under the Gujarat Private Universities Act, 2009.

The university offers Diploma, Undergraduate, Postgraduate and Doctoral programmes across Engineering, Technology, Management, Commerce, Design, Law, Science, Humanities, Healthcare and Allied disciplines. More than 25,000 students study across its constituent colleges and institutes.

In May 2025, Silver Oak University received Grade A accreditation from the National Assessment and Accreditation Council (NAAC). The university describes itself as the youngest university in Ahmedabad to achieve this distinction.

2. Objective:

- To identify important information and data assets.
- To identify potential threats to university data.
- To identify vulnerabilities that could be exploited by threats.
- To identify appropriate security controls.
- To improve the overall data-security posture of the university.
- To protect the confidentiality, integrity, and availability of university information.

3. Scope:

The scope of this risk assessment is limited to data and information security within Silver Oak University. The university provides data-centric programs across engineering, computer applications, and management streams. Data science and tech graduates from Silver Oak have consistent pathways into local and multinational corporate hubs.

4. Data Assets and Their Importance:

Data Asset	Example Information	Importance
Student Personal Data	Name, DOB, contact details, address, ID	Very High
Academic Data	Marks, grades, attendance, courses	Critical
Examination Data	Question papers, answer data, results	Critical
Financial Data	Fees, transactions, payment records	Very High
Faculty Data	Personal and professional information	High
HR Data	Employee records, salary and documents	Very High
Login Credentials	Username, passwords, authentication information	Critical
Research Data	Research documents, datasets, intellectual property	High
Backup Data	Copies of databases and important documents	Critical

5. Risk Identification:

Risk identification is the process of identifying possible events that may negatively affect university data.

- a) **Unauthorized Access:** Unauthorized individuals may gain access to student, faculty, financial, or administrative information through compromised credentials or inadequate access controls.
- b) **Data Breach:** Sensitive student or employee information could potentially be exposed, copied, or transferred without authorization.
- c) **Phishing and Credential Theft:** Students, faculty, or employees may receive fraudulent emails or messages designed to obtain their usernames, passwords, or other sensitive information.
- d) **Ransomware:** Malware could encrypt university files and databases and make critical information unavailable.
- e) **Accidental Data Loss:** Important information could be deleted accidentally by students, faculty, administrators, or IT personnel.
- f) **Insider Threat:** An authorized user could intentionally or accidentally access, modify, copy, or disclose information beyond their legitimate requirement.
- g) **Weak Authentication:** Weak passwords or absence of multi-factor authentication may increase the possibility of account compromise.
- h) **Excessive User Privileges:** Users may receive more access to information than is necessary for their role.
- i) **Data Modification:** Unauthorized modification of academic, examination, attendance, or financial records could affect data integrity.
- j) **Inadequate Backup:** If important data is not backed up properly, recovery may be difficult after ransomware, hardware failure, accidental deletion, or other incidents.
- k) **Malware Infection:** Malicious software introduced through email attachments, downloads, USB devices, or compromised websites could affect university systems.

6. Risk Analysis:

Risk	Likelihood	Impact	Risk Score	Risk Level
Student data breach	4	5	20	Critical
Examination data leakage	3	5	15	High
Ransomware/data encryption	4	5	20	Critical
Phishing and credential theft	4	4	16	High
Unauthorized employee access	3	4	12	High
Academic data modification	3	5	15	High
Financial data exposure	3	5	15	High
Accidental data deletion	3	4	12	High
Weak authentication	4	4	16	High
Research data theft	3	4	12	High
Malware infection	3	4	12	High
Backup failure	3	5	15	High
Website/data-system compromise	3	3	9	Medium

7. Mitigation Report:

Risk mitigation involves implementing controls to reduce the probability or impact of identified risks.

Problem	Solution
Weak student/faculty passwords	Implement strong password policies and password protection controls

No multi-factor authentication	Implement MFA for administrators, faculty, employees and sensitive systems
Excessive user permissions	Implement Role-Based Access Control (RBAC) and least-privilege access
Unauthorized access to student data	Restrict database access according to user roles and responsibilities
Phishing attacks	Conduct regular cybersecurity awareness and phishing-awareness training
Ransomware	Deploy endpoint protection, patch systems regularly and maintain offline/isolated backups
Accidental data deletion	Implement automated backups and data-recovery procedures
Examination data leakage	Restrict access to examination information and maintain detailed access logs
Academic record modification	Implement authorization controls, audit trails and approval mechanisms
Financial data exposure	Apply encryption, access control and secure authentication to financial systems
Outdated software	Establish a regular patch and vulnerability-management program
Malware infection	Use endpoint security, email filtering and malware detection mechanisms
Insider misuse	Apply least privilege, access reviews and user activity monitoring
Backup failure	Follow a structured backup strategy and periodically test restoration
Unauthorized database access	Restrict database connectivity and monitor database activity

8. Risk Register Report:

The risk register provides a summarized record of identified risks, their impact, risk level, controls, and current status.

ID	Risk	Impact	Risk Level	Control	Status
R01	Student personal data breach	Privacy, reputation and compliance impact	Critical	Encryption, RBAC, MFA, monitoring	Open
R02	Examination data leakage	Examination integrity and reputation	High	Restricted access, encryption, audit logs	Open
R03	Ransomware attack	Data unavailability and operational disruption	Critical	Endpoint security, patching, isolated backups	In Progress
R04	Phishing and credential theft	Account compromise and data exposure	High	MFA, awareness training, email filtering	In Progress
R05	Excessive user privileges	Unauthorized data access	High	RBAC, least privilege, periodic access review	Open

R06	Academic record modification	Incorrect marks/results and loss of trust	High	Authorization, audit trails, approvals	Open
R07	Financial data exposure	Financial and privacy impact	High	Encryption, MFA, access controls	Open
R08	Accidental data deletion	Loss of important information	High	Automated backup and recovery	In Progress
R09	Weak authentication	Unauthorized account access	High	Strong passwords and MFA	In Progress
R10	Research data theft	Loss of intellectual property	High	Access control, encryption and monitoring	Open
R11	Malware infection	Data corruption and system disruption	High	Endpoint protection and email security	In Progress
R12	Backup failure	Permanent or prolonged data loss	High	3-2-1 backup strategy and recovery testing	Open
R14	Insider data misuse	Confidentiality and integrity impact	High	Least privilege, logging and monitoring	Open

R1 5	Outdated software	System compromise and data breach	Medium	Patch management and vulnerability assessment	In Progress
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9. Conclusion:

The risk assessment demonstrates that data is one of the most important information assets within a modern private university environment. Student personal information, academic records, examination data, financial information, employee records, research information, credentials, and backup data require appropriate security controls.

The major potential risks identified in this assessment include **data breaches, unauthorized access, phishing, ransomware, excessive privileges, insider misuse, academic data manipulation, financial data exposure, accidental data loss, malware, and backup failures.**

The university can reduce these risks through a combination of **Multi-Factor Authentication, Role-Based Access Control, encryption, least privilege, regular backups, patch management, endpoint protection, security awareness training, monitoring, audit logging, vulnerability assessments, and incident-response planning.**

Risk assessment should not be treated as a one-time activity. University systems, users, applications, and threats continuously change. Therefore, the risk register should be reviewed periodically and updated whenever new systems, applications, data types, threats, or vulnerabilities are introduced.

Overall, a strong data-security and risk-management program can help protect university information assets while maintaining the **confidentiality, integrity, and availability (CIA)** of critical data.